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让蚂蚁变僵尸的真菌

The Fungus That Turns Ants into Zombies

蚂蚁为何会咬住叶片不松口?

英文原文

Deep in a Thai **rainforest**, a **carpenter** ant breaks from its **colony**'s **column** in the **lofty canopy**. **Swaying** on **unsteady** legs, it **wanders downward**, **as if intoxicated**, into **leafy undergrowth**. Around **midday** it **clenches** its **jaws** on a major **vein** underneath a leaf, about 25 centimetres above the ground.

By **dusk** the insect is **lifeless**, yet the bite holds **rigidly**. A fungus has destroyed its jaw **muscles**, so they cannot **loosen**. Two or three days later a **fungal** stalk pushes from its head and releases **infectious** spores on to the ground below, where foraging ants **tread**.

The **invader**, Ophiocordyceps unilateralis, is **fussy** about **location**. David Hughes and colleagues found the **bites** mostly on the **north-west** side of plants, where the air is **humid** and the temperature suits the fungus. When researchers moved infected ants higher, the fungus grew **abnormally**. Placed on the ground, the bodies **disappeared**.

In 2017 a Penn State team examined infected ants with **three-dimensional microscope** imaging. Fungal cells had formed an **intertwined lattice** around muscle fibres in the legs and jaw, a picture that **evokes** the strings of a **puppet**. Yet the fungus had not **infiltrated** the brain.

A Planet Earth episode, with David Attenborough's **narration**, showed one such ant. It helped **inspire** The Last of Us, a 2013 video game whose **makers envisioned** the parasite jumping to humans. The **hypothetical scenario** gave **gamers** a **fictional** nightmare.

The real parasite's roots run far deeper, into the **geological** record. A 48-million-year-old **fossil** leaf from Messel, in Germany, bears the same **characteristic** bite **scars**.

中文译文

在泰国雨林深处，一只木匠蚁脱离了它的蚁群队列，在高耸的树冠上（前行）。摇摇晃晃，腿脚不稳，它向下游荡，好像喝醉了一样，进入枝叶茂密的林下植被。正午前后，它将双颚紧咬在一片叶子下方的一条主叶脉上，离地面约25厘米。

到黄昏时，这只昆虫已无生命，那一口却仍牢牢地咬着。一种真菌摧毁了它的颚部肌肉，因此它们无法松开。两三天后，一根真菌菌柄从它头部顶出，将具有传染性的孢子释放到下方的地面上，那里有觅食的蚂蚁行走。

这个入侵者，也就是偏侧蛇虫草菌，对位置很挑剔。戴维·休斯和同事发现，这些咬合点大多在植物的西北侧，那里空气潮湿，且温度适宜真菌。研究人员把受感染的蚂蚁移到更高处时，真菌长得很反常。放在地面上，尸体则不见了。

2017年，宾夕法尼亚州立大学的一个团队检查了受感染的蚂蚁，用的是三维显微镜成像。真菌细胞形成了相互交织的网格，围在腿部和颚部的肌肉纤维周围，一幅使人想起木偶的提线的画面。不过，真菌尚未渗入大脑。

《地球脉动》的一集，伴随着戴维·阿滕伯勒的解说，就展示了这样一只蚂蚁。它帮助启发了2013年的电子游戏《最后生还者》，游戏的制作者们设想这种寄生菌跳到了人类身上。这个假想的情景给玩家带来了一场虚构的噩梦。

现实中的寄生菌，其根源要深得多，一直延伸到地质记录之中。一片4800万年前的叶子化石，出土于德国梅塞尔，上面带有同样典型的咬合疤痕。

词汇与表达

rainforest n.
雨林

carpenter n.
木匠

colony n.
群
its colony's column

column n.
队列

lofty adj.
高耸的

canopy n.
树冠
in the lofty canopy

Swaying v.
摇摇晃晃

unsteady adj.
不稳
on unsteady legs

wanders v.
游荡
it wanders downward

downward adv.
向下

as if 连词短语
好像

intoxicated adj.
喝醉了
as if intoxicated

leafy adj.
枝叶茂密的

undergrowth n.
林下植被
into leafy undergrowth

midday n.
正午

clenches v.
紧咬
it clenches its jaws

jaws n.
双颚

vein n.
叶脉
a major vein underneath a leaf

dusk n.
黄昏
By dusk

lifeless adj.
无生命

rigidly adv.
牢牢地
the bite holds rigidly

muscles n.
肌肉

loosen v.
松开

fungal adj.
真菌

infectious adj.
具有传染性的
releases infectious spores

tread v.
行走
where foraging ants tread

词汇与表达（续）

invader n.
入侵者

fussy adj.
挑剔
is fussy about location

location n.
位置

bites n.
咬合点

north-west n.
西北

humid adj.
潮湿
the air is humid

abnormally adv.
反常
the fungus grew abnormally

disappeared v.
不见了

three-dimensional adj.
三维

microscope n.
显微镜

intertwined adj.
相互交织的
an intertwined lattice

lattice n.
网格
an intertwined lattice

evokes v.
使人想起
a picture that evokes

puppet n.
木偶

infiltrated v.
渗入
had not infiltrated the brain

narration n.
解说

inspire v.
启发了

makers n.
制作者

envisioned v.
设想

hypothetical adj.
假想的
The hypothetical scenario

scenario n.
情景
The hypothetical scenario

gamers n.
玩家

fictional adj.
虚构的

geological adj.
地质

fossil n.
化石
fossil leaf from Messel

characteristic adj.
典型的

scars n.
疤痕

文章简述

- 感染：泰国木匠蚁感染一种寄生真菌后离群，正午咬住离地约25厘米的叶背主脉，到黄昏死去。
- 传播：真菌破坏控制咬合的肌肉，使下颚无法松开，蚂蚁死后仍咬紧叶子；两三天后，蚁尸头部长出真菌细柄，向蚂蚁觅食的地面释放孢子，也就是可传播感染的微小颗粒。
- 环境：咬痕多在温湿度适宜的植物西北侧；把感染蚂蚁移高，真菌生长反常，把蚁尸放到地面，尸体则消失。
- 机制：2017年三维成像显示，真菌在腿和颚部肌肉周围织成网络，却没有侵入蚂蚁大脑。
- 边界：4800万年前的叶化石已有同类咬痕；记录这种现象的纪录片启发了游戏《最后生还者》，但游戏中真菌感染人类的情节是虚构的。

禁区里的生机

Life in the Forbidden Zone

禁区里动物变多，就能说明辐射无害吗？

英文原文

Chernobyl is **synonymous with disaster**, yet it holds a **startling** amount of **wildlife**. Early on 26 April 1986, one of its **reactors** exploded in Soviet Ukraine, **dispersing radioactive fallout** across Europe. On 27 April, nearby Pripyat **was emptied of** about 45,000 **inhabitants** within hours. A zone some 30 kilometres in **radius** became **off-limits**, and Belarus **fenced off adjoining terrain**.

In October 2015, a study in **Current Biology** combined **aerial** surveys with **footprint** counts on the Belarusian side. **Elk**, **roe deer**, red deer, and wild **boar** were **comparable** in **density to** those in four **reserves** free of **contamination**, and wolves were over seven times as **numerous**. Earlier **helicopter** counts showed **elk**, **roe deer**, and **boar surging** in the **post-disaster** decade.

Later **fieldwork bolstered** these findings. In April 2016, camera traps **snapped** wolves, boar, and raccoon dogs near the most **hazardous** spots, with no sign of **suppressed** numbers. The **probable** reason is **mundane**: the **absence** of **agriculture**, **forestry**, and hunting.

Smaller **organisms** fared worse in some studies. One, in March 2007, found fewer **nesting** birds, and fewer species, where radiation was more **intense**. Another, in March 2009, found **pollinators** and spiders less **plentiful** where **exposure** was greater. A third, in March 2014, found leaves **decomposing** about 40 per cent less at the most **polluted** sites, **impeding** the return of nutrients to the soil.

Counting heads cannot **guarantee** the health of **individual** animals, and **how much damage** low but **continual** doses **inflict** remains **contentious**, as studies **clash over** methods. The **burgeoning herds** may **trumpet** the **dividend** of human **departure**, but the **sluggish rot** of fallen leaves keeps a quieter **ledger** of radiation's cost.

中文译文

切尔诺贝利，（虽然）是灾难的代名词，但是这里却容纳着数量惊人的野生动物。1986年4月26日凌晨，它的一座反应堆爆炸于苏联乌克兰，扩散放射性的尘降物遍及整个欧洲。在4月27日，附近的普里皮亚季被清空了约45,000名居民，发生在数小时内。一片半径约30公里的区域变成了禁区，白俄罗斯也用栅栏隔开了相邻的地带。

2015年10月，《当代生物学》上的一项研究结合了空中的调查与足迹计数，这些数据都在白俄罗斯一侧。驼鹿、狍、马鹿和野猪在数量密度方面，相当于四个未受污染的保护区里的同类，而狼的数量非常多，是那些保护区里的同类的七倍之多。更早的直升机计数（空中盘点）显示，驼鹿、狍和野猪在灾后的那十年里数量激增。

后来的实地考察佐证了这些发现。2016年4月，相机陷阱在最危险的几个地点附近，拍到了狼、野猪和貉，没有迹象表明动物的数量被抑制。可能的原因很平常：没有农耕、伐木和狩猎。

在一些研究中，更小的生物过得更差。一项研究于2007年3月发现了更少的筑巢的鸟类，以及更少的种类，在这里辐射更加强烈。另一项研究于2009年3月发现，传粉昆虫和蜘蛛没那么丰富/数量众多，在这些地方，辐射接触更大。第三项研究于2014年3月发现，在最严重被污染的地点，落叶的分解少了约40%，阻碍了养分回归土壤。

仅靠清点动物的数量无法确保动物个体就是健康的；并且较低但是却一直持续的辐射剂量究竟会（对生物）造成多大伤害，目前仍然存在争议，因为各项研究在方法上存在冲突。迅速增长的兽群或许在宣扬人类离开所带来的红利，但落叶迟缓的腐烂，记着一笔更安静的账：辐射的代价。

词汇与表达

synonymous with adj.
代名词
is synonymous with disaster

disaster n.
灾难

startling adj.
惊人的

wildlife n.
野生动物

reactors n.
反应堆

dispersing v.
扩散
dispersing radioactive fallout

radioactive adj.
放射性的
radioactive fallout across Europe

fallout n.
尘降物
radioactive fallout across Europe

was emptied of v.
被清空了

inhabitants n.
居民
about 45,000 inhabitants

radius n.
半径

off-limits adj.
禁区

fenced off v.
用栅栏隔开了

adjoining adj.
相邻的

terrain n.
地带

Current Biology n.
《当代生物学》

aerial adj.
空中的

footprint n.
足迹

Elk n.
驼鹿

roe deer n.
狍

boar n.
野猪

comparable to adj.
相当……于

词汇与表达（续）

density n.
密度
comparable in density

reserves n.
保护区

contamination n.
污染
free of contamination

numerous adj.
数量非常多

helicopter n.
直升机

surging v.
激增

post-disaster adj.
灾后的

fieldwork n.
实地考察
Later fieldwork bolstered

bolstered v.
佐证了
fieldwork bolstered these findings

snapped v.
拍到了

hazardous adj.
危险的
the most hazardous spots

suppressed adj.
被抑制

probable adj.
可能的

mundane adj.
平常

absence n.
没有
the absence of agriculture

agriculture n.
农耕

forestry n.
伐木

organisms n.
生物
Smaller organisms fared worse

nesting adj.
筑巢的

intense adj.
强烈

pollinators n.
传粉昆虫

词汇与表达（续）

plentiful adj.
丰富/数量众多

exposure n.
接触
where exposure was greater

decomposing v.
分解
leaves decomposing about 40 per cent

polluted adj.
被污染的

impeding v.
阻碍了
impeding the return of nutrients

guarantee v.
确保
guarantee the health

individual adj.
个体

how much damage 名词短语
多大伤害

continual adj.
一直持续的
low but continual doses

inflict v.
造成
low but continual doses inflict

contentious adj.
争议
remains contentious

clash over v.
在……上存在冲突

burgeoning adj.
迅速增长的

herds n.
兽群

trumpet v.
宣扬

dividend n.
红利
the dividend of human departure

departure n.
离开

sluggish adj.
迟缓的
the sluggish rot of fallen leaves

rot n.
腐烂

ledger n.
账

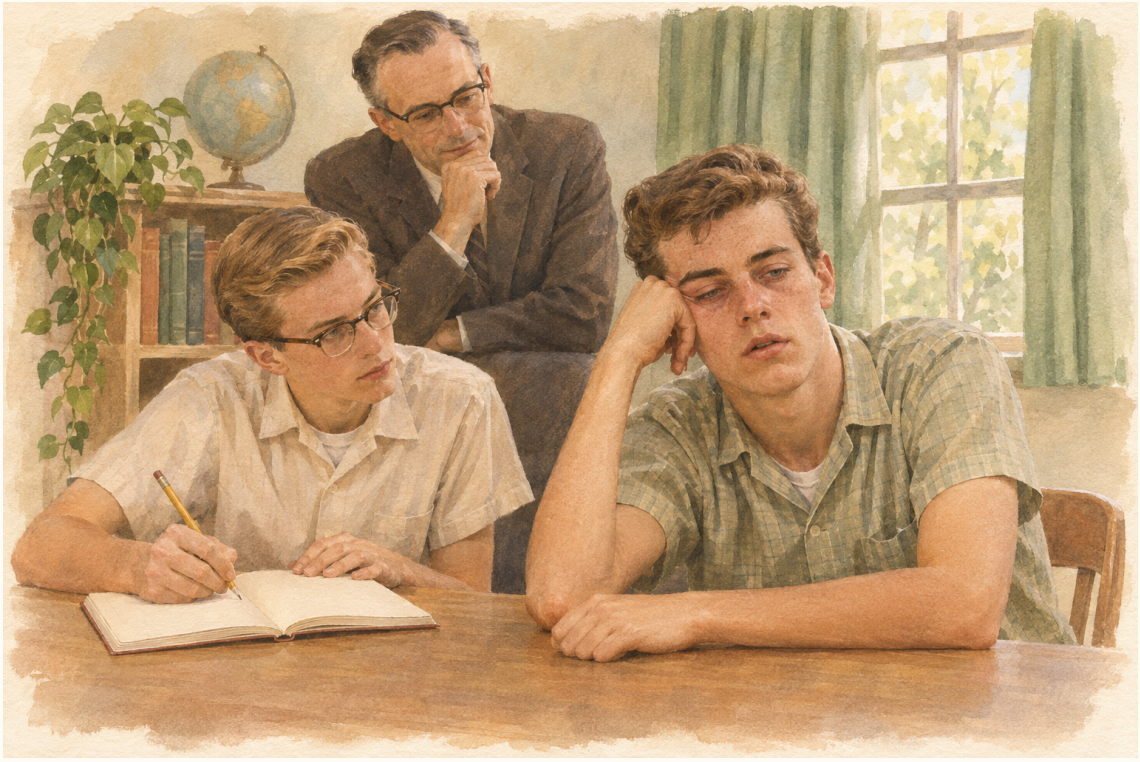
文章简述

- 灾后：1986年切尔诺贝利核事故后，居民撤离、周边封禁，受污染地区却出现大量野生动物。
- 繁盛：白俄罗斯一侧多种兽类密度与四个未污染保护区相当，狼的数量为对照区的七倍多。
- 原因：可能的解释是农耕、伐木和狩猎消失，人类活动减少给野生动物带来了生存空间。
- 代价：部分研究发现高辐射处筑巢鸟数量及鸟种较少，传粉昆虫和蜘蛛也少；污染最重地点的落叶分解明显变慢，养分更难回到土壤。
- 限度：动物多不等于个体健康；长期低剂量辐射造成多大伤害仍有争议，不能据此说辐射无害。

23

十一天不睡觉的少年

Eleven Days Without Sleep



十一天不睡觉，身体会怎样？

英文原文

On 28 December 1963 Randy Gardner, a 17-year-old science **enthusiast** at a San Diego high school, set out to **forgo slumber** for a **regional** science fair. Two **companions diligently jolted** him **awake** whenever his **alertness waned** and logged his **mood**. The project, a **voluntary endeavour**, went on to **take first place**.

William Dement, a sleep researcher at Stanford, read press **coverage** of the attempt and was **intrigued**. He drove to San Diego and **assisted** John J. Ross, a Navy researcher, **in** monitoring the **sleepless** boy **closely**. The two **investigators attentively** noted the **fluctuations** in his state as the **deprivation progressively deepened**.

The **consequences** soon **mounted**. Gardner grew **moody**, lost **concentration**, and found his **short-term** memory **impaired**. He became **suspicious**, at times **paranoid**, and began to **hallucinate**. On the fourth day he was **gripped by** the **delusion** that he was a professional American football player.

Gardner ended the **wearying** attempt on 8 January 1964, after 264.4 hours, about 11 days and 25 minutes. He then slept **soundly** for about 14 hours and woke apparently **refreshed**. Researchers **probed** the case in a 1965 **scholarly** paper, which helped **highlight** microsleeps, **involuntary lapses into** sleep lasting a few seconds. Gardner may have had such **momentary bouts** without noticing.

In 1997 Guinness World Records **discontinued** monitoring such attempts because of the health **perils**, and others have since **alleged** longer **stints** awake. In a 2017 interview Gardner said severe **insomnia** had **tormented** him for about a decade, and he himself **pinned the blame on** the experiment. The boy who had seemed to **bounce back from exhaustion** with one long rest would, decades later, **barely doze off**.

中文译文

1963年12月28日，圣迭戈一名17岁的高中科学爱好者兰迪·加德纳，为了参加一场地区性的科学展，开始尝试放弃睡眠。两名同伴尽心尽力，每当他的警觉减弱，就让他猛然清醒过来，并记录了他的情绪。这个项目，一次自愿的尝试，继而夺得了第一名。

威廉·德门特，斯坦福大学的睡眠研究者，读到了媒体对这次尝试的报道，并且非常感兴趣。他驾车前往圣迭戈，协助约翰·J·罗斯，一位海军研究员，去密切监视这个不眠的少年。两位研究人员专心地记下了他状态的种种波动，其间睡眠剥夺逐步加深。

后果很快便开始不断加剧。加德纳变得喜怒无常，注意力难以集中，他的短时记忆也受损了。他变得多疑，有时甚至偏执/妄想性的，还开始出现幻觉。第四天，他被一种妄想/错觉攫住了，以为自己是一名美式橄榄球职业球员。

加德纳于1964年1月8日结束了这场令人疲惫的尝试，共历时264.4小时，约合11天零25分钟。随后他酣畅地/深度地睡了约14小时，醒来时似乎神清气爽/恢复精神的。研究者们在1965年的一篇学术论文中探究了/深入研究这一案例，该论文有助于凸显微睡眠——指不由自主地陷入持续几秒钟短暂睡眠的状态。加德纳可能经历过这种短暂的发作而未曾察觉。

1997年，吉尼斯世界纪录因健康风险停止了监测此类尝试；此后，其他人声称/宣称有过更长的清醒时段。在2017年的一次采访中，加德纳说严重的失眠折磨了他约十年，他本人把它归咎于这次实验。那个曾看似只要睡个长觉，就能从精疲力竭中迅速恢复的少年，几十年后，却几乎连打个盹都费劲。

词汇与表达

enthusiast n.
爱好者
a 17-year-old science enthusiast

forgo v.
放弃
to forgo slumber

slumber n.
睡眠

regional adj.
地区性的

companions n.
同伴

diligently adv.
尽心尽力

jolted awake v.
让……猛然清醒过来

alertness n.
警觉
his alertness waned

waned v.
减弱
his alertness waned

mood n.
情绪

voluntary adj.
自愿的
a voluntary endeavour

endeavour n.
尝试
a voluntary endeavour

take first place 动词短语
夺得了第一名

coverage n.
报道

intrigued adj.
非常感兴趣
was intrigued

assisted in v.
协助

sleepless adj.
不眠的

closely adv.
密切

investigators n.
研究人员

attentively adv.
专心地

fluctuations n.
波动
the fluctuations in his state

deprivation n.
剥夺
the deprivation progressively deepened

progressively adv.
逐步

deepened v.
加深

consequences n.
后果

mounted v.
不断加剧

moody adj.
喜怒无常

concentration n.
注意力
lost concentration

short-term adj.
短时

词汇与表达（续）

impaired adj.
受损了
short-term memory impaired

suspicious adj.
多疑

paranoid adj.
偏执/妄想性的
at times paranoid

hallucinate v.
出现幻觉
began to hallucinate

gripped by v.
被……攫住了

delusion n.
妄想/错觉
gripped by the delusion

wearying adj.
令人疲惫的

soundly adv.
酣畅地/深度地

refreshed adj.
神清气爽/恢复精神的

probed v.
探究了/深入研究

scholarly adj.
学术

highlight v.
凸显

involuntary adj.
不由自主地
involuntary lapses into sleep

lapses into n.
陷入

momentary adj.
短暂的

bouts n.
发作

discontinued v.
停止了
discontinued monitoring such attempts

perils n.
风险
because of the health perils

alleged v.
声称/宣称

stints n.
时段

insomnia n.
失眠
severe insomnia

tormented v.
折磨了
had tormented him for about a decade

pinned the blame on v.
把……归咎于
pinned the blame on the experiment

bounce back from v.
从……中迅速恢复
bounce back from exhaustion

exhaustion n.
精疲力竭
bounce back from exhaustion

barely adv.
几乎

doze off v.
打个盹

文章简述

- 挑战：1963年底，17岁高中生兰迪·加德纳为参加科学展尝试不睡觉；同伴帮他保持清醒，睡眠研究者随后加入监测。
- 损害：他逐渐情绪不稳，注意力和短时记忆受损，出现多疑和幻觉，甚至误以为自己是职业橄榄球员。
- 休息：264.4小时、约11天后，他结束挑战；酣睡约14小时，醒来时似乎恢复了精神。
- 疑点：他可能曾不自觉地睡着几秒，这叫微睡眠，本人未必察觉；吉尼斯1997年因健康风险停止记录此类挑战。
- 晚年：2017年，他说严重失眠已折磨自己约十年，并认为是当年实验造成的；文章没有证明两者有因果关系。

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集装箱：重塑世界的铁盒子

The Steel Box That Remade the World

一个普通铁箱，怎样改变了全球货运？

英文原文

On 26 April 1956 a **converted** Second World War **tanker**, the Ideal X, **slipped out of** Port Newark, New Jersey, for Houston. **Clamped** to her deck were 58 **containers**, **trailer** bodies **detached from** their **wheels** and **chassis**. Few **spectators** would have taken so **unremarkable** a **cargo** for a **landmark**.

At the time, **freight** travelled **loose**. **Gangs** of twenty or more **dockworkers** carried **sacks**, **barrels**, and **crates** piece by piece. It was **strenuous drudgery**. Ships could spend **up to** half their time **berthed**, and **theft** was **rampant**. In 1956 **loading** loose cargo cost about $5.83 a ton; loading the Ideal X cost, **by** the company's **reckoning**, about 16 cents.

Containers were not new. The **hurdle** was **consensus**: **rival** firms' boxes had **incompatible dimensions** and **latches**. An **international standards** committee formed in 1961. In January 1963 Sea-Land, the Ideal X's renamed operator, let the committee use its **patents** free of **royalties**, including those on corner **fittings** for **hoisting**. Lifted by **crane**, a box could be **transferred smoothly** between ship, **lorry**, and train.

This **efficiency** came **at the expense of livelihoods**. New York's **waterfront employment shrank** by about three quarters between 1963 and 1976. Much of the cargo crossed the harbour to a new **terminal** at Elizabeth, **inaugurated** in August 1962. Most of London's **historic docks** closed one after another from 1967 to 1980, while Felixstowe, on the Suffolk coast, **expanded**.

The **rewards** were **substantial**. A study published in the Journal of International **Economics** in 2016 estimated that containerisation had lifted trade between **industrialised** countries by a **cumulative** 700 per cent over 20 years. **Tireless** cranes now lift **identical** containers into **orderly**, **immense stacks**.

中文译文

1956年4月26日，一艘改造过的二战油轮“理想X号”，悄然驶离新泽西州纽瓦克港，开往休斯敦。固定在她甲板上的是58个集装箱，其实就是从车轮和底盘上拆卸下来的挂车车厢。极少数旁观者会把一批如此平淡无奇的货物当作一个里程碑。

那时，货物都是散装运输。一帮二十多名或更多的码头工人，一件件地搬运麻袋、桶和板条箱。这是繁重的/费力的苦差事。船舶可能会花多达一半的时间停泊在港，而且偷窃猖獗。1956年，装载散货每吨花费约5.83美元；据公司估算，装载理想X号花费约16美分。

集装箱并不新鲜。障碍在于共识：竞争对手公司的箱子有互不兼容的尺寸和门闩/锁扣。1961年，一个国际标准委员会成立。1963年1月，海陆公司，即理想X号更名后的运营方，允许该委员会使用其专利，不收取专利费，包括用于吊装的角配件上的那些专利。由吊机吊起，一只箱子便能在船舶、卡车和火车之间顺畅地转运。

这种效率是以生计为代价换来的。纽约海滨区的就业人数在1963年至1976年间缩减了约四分之三。大量货物穿过港湾，前往伊丽莎白一处于1962年8月启用的新码头/港区。1967年到1980年，伦敦历史悠久的码头一个接一个地关闭，而位于萨福克海岸的费利克斯托则扩张了。

回报相当可观。一项2016年发表于《国际经济学杂志》的研究估计，集装箱化在20年间使工业化国家之间的贸易提高了累计700%。如今，不知疲倦的吊机把一模一样的集装箱吊起，码成整齐的、巨大的箱垛。

词汇与表达

converted adj.
改造过的

tanker n.
油轮

slipped out of v.
悄然驶离

Clamped v.
固定

containers n.
集装箱

trailer n.
挂车

detached from v.
从……上拆卸下来的

wheels n.
车轮

chassis n.
底盘

spectators n.
旁观者

unremarkable adj.
平淡无奇的

cargo n.
货物
so unremarkable a cargo

landmark n.
里程碑
a cargo for a landmark

freight n.
货物
freight travelled loose

loose adj.
散装

Gangs n.
帮

dockworkers n.
码头工人

sacks n.
麻袋

barrels n.
桶

crates n.
板条箱

strenuous adj.
繁重的/费力的

drudgery n.
苦差事
strenuous drudgery

up to prep.
多达

词汇与表达（续）

berthed v.
停泊

theft n.
偷窃

rampant adj.
猖獗
theft was rampant

loading n.
装载

by reckoning phr.
据……估算

hurdle n.
障碍
The hurdle was consensus

consensus n.
共识
The hurdle was consensus

rival adj.
竞争对手

incompatible adj.
互不兼容的
incompatible dimensions and latches

dimensions n.
尺寸
incompatible dimensions and latches

latches n.
门闩/锁扣

international standards n.
国际标准

patents n.
专利
use its patents free of royalties

royalties n.
专利费
use its patents free of royalties

fittings n.
配件

hoisting v.
吊装

crane n.
吊机
Lifted by crane

transferred v.
转运
could be transferred smoothly

smoothly adv.
顺畅地

lorry n.
卡车

词汇与表达（续）

efficiency n.
效率
This efficiency came

at the expense of n.
以……为代价
at the expense of livelihoods

livelihoods n.
生计
at the expense of livelihoods

waterfront n.
海滨区

employment n.
就业人数
waterfront employment shrank

shrank v.
缩减了

terminal n.
码头/港区

inaugurated v.
启用
inaugurated in August 1962

historic adj.
历史悠久的

docks n.
码头

expanded v.
扩张了

rewards n.
回报

substantial adj.
可观
The rewards were substantial

Economics n.
经济学

industrialised adj.
工业化

cumulative adj.
累计
a cumulative 700 per cent

Tireless adj.
不知疲倦的

identical adj.
一模一样的
lift identical containers

orderly adj.
整齐的

immense adj.
巨大的

stacks n.
箱垛

文章简述

- 启航：过去码头逐件搬散货，船常久停；1956年，理想X号载58个集装箱启航。
- 降本：据运营方估算，该船每吨装船费约0.16美元，远低于当时散货的5.83美元。
- 互通：各家箱型原不兼容；运营方海陆公司免费开放专利，推动标准统一，箱子可在船、卡车和火车间转运。
- 转移：货运转向新港，老码头和工人生计受冲击；1963至1976年，纽约滨水区就业缩减约四分之三。
- 回报：2016年一项研究估计，集装箱化在20年间使工业化国家间贸易累计增加约700%。

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永不沉没的神话

The Myth of the Unsinkable

守法的泰坦尼克号，为何仍救不了所有人？

英文原文

On 10 April 1912 the **opulent liner** Titanic left Southampton on her **maiden voyage**. A 1911 **trade journal** had **dubbed** her and her sister ship "**practically unsinkable**". At about 11.40pm on 14 April she **collided with** an **iceberg**; by 2.20am she had **vanished** beneath the waves, and the **death toll** would **surpass** 1,500.

Folklore blames **arrogance**. Yet the Titanic was **compliant with** the law. Under 1894 rules, Britain's **maritime regulator pegged** lifeboat numbers **to gross tonnage** (volume, not weight) on a **scale** that **topped out at** 16 boats for 10,000 tons or more. At about 46,000 tons she carried 20, **exceeding** the **requirement**. Their **capacity** was 1,178, **roughly** half the 2,200 aboard: **sufficient** in law, **inadequate** at sea.

Within radio range, the steamer Californian kept an **equally lawful** routine. Her **sole wireless operator**, Cyril Evans, was **on the payroll of** the Marconi company; when he slept, so did the radio. Around 11.30pm he **switched off** his set. **Hindsight** may call that **negligence**, yet no rule **obliged** him **to** stay **vigilant**, so the Titanic's **frantic distress calls** went **unheard** there.

Between April and July 1912, inquiries in Washington and London heard **testimony** from **survivors** and the Californian's crew. That July, American **legislation** made **round-the-clock** radio watch **compulsory**. In January 1914 **delegates** of 13 nations signed the first Safety of Life at Sea **convention**, **stipulating** lifeboat seats for all, **continuous** radio watch, and an ice **patrol**. The First World War **stalled** the **treaty**; the patrol went ahead **regardless**.

The **doomed** liner sank within three hours. The **outdated** rules she sailed under had **stayed afloat** since 1894; it took her dead to **discredit** them.

中文译文

1912年4月10日，豪华的邮轮泰坦尼克号离开了南安普敦，开始了她的处女航。1911年，一家行业期刊把她和她的姊妹船称为“几乎不会沉没的”。4月14日晚约11时40分，她和一座冰山相撞了；到次日凌晨2时20分，她已消失在海浪之下，遇难人数最终将超过1,500。

坊间流传的说法把灾难归咎于傲慢。然而泰坦尼克号符合法规。按照1894年的规则，英国的海事监管机构把救生艇数量与总吨位（体积，而非重量）挂钩，在一个等级表上/等级体系中，（其数量）最高达（只能）到16艘，针对10,000吨及以上（的船）。她约有46,000吨，带了20艘，超出了要求。它们的容量是1,178人，大约是船上2,200人的一半：按法律足够，到了海上却不足。

在无线电通讯范围内，加州人号轮船遵循的是一套同样合法的值班惯例。她唯一的无线电操作员西里尔·埃文斯，受雇于马可尼公司；他一睡下，电台也一样睡了。晚上11时30分左右，他关掉了自己的设备。事后看来，这或许可称为疏忽，然而没有哪条规则强制他保持警觉，所以泰坦尼克号发狂似的求救呼叫，在那里无人听见。

1912年4月至7月，华盛顿和伦敦的调查听取了幸存者以及加州人号船员的证词。同年7月，美国的一项立法将全天候无线电值守定为强制性的（义务）。1914年1月，13国的代表签署了第一部《海上人命安全公约》，规定救生艇座位要够所有人乘坐，电台要不间断值守，还要开展冰情巡逻。第一次世界大战让这项条约暂缓搁置了；然而冰情巡逻不管怎样依然进行。

这艘在劫难逃的邮轮不到三小时就沉没了。她航行时所遵循的那套过时的规则，却自1894年起一直没有沉没；直到她的乘客葬身大海，它们才信誉扫地。

词汇与表达

opulent adj.
豪华的
the opulent liner Titanic

liner n.
邮轮

maiden voyage adj.
处女航
on her maiden voyage

trade journal 名词短语
行业期刊

dubbed v.
把……称为
had dubbed her and her sister ship

practically adv.
几乎

unsinkable adj.
不会沉没的

collided with v.
和……相撞了
she collided with an iceberg

iceberg n.
冰山

vanished v.
消失

death toll n.
遇难人数
the death toll would surpass 1,500

surpass v.
超过

Folklore n.
坊间流传的说法

arrogance n.
傲慢
Folklore blames arrogance

compliant with adj.
符合
was compliant with the law

maritime adj.
海事

regulator n.
监管机构
Britain's maritime regulator

pegged to v.
把……与……挂钩

gross tonnage adj.
总吨位
gross tonnage (volume, not weight)

scale n.
等级表……等级体系

topped out at 动词短语
最高达……到

exceeding v.
超出了

requirement n.
要求

capacity n.
容量
Their capacity was 1,178

roughly adv.
大约

sufficient adj.
足够
sufficient in law

inadequate adj.
不足
inadequate at sea

equally adv.
同样

词汇与表达（续）

lawful adj.
合法的

sole adj.
唯一的

wireless adj.
无线电

operator n.
操作员

on the payroll of n.
受雇于

switched off 动词短语
关掉了

Hindsight n.
事后看来
Hindsight may call that negligence

negligence n.
疏忽
Hindsight may call that negligence

obliged to v.
强制

vigilant adj.
警觉
no rule obliged him to stay vigilant

frantic adj.
发狂似的

distress calls n.
求救呼叫

unheard adj.
无人听见

testimony n.
证词
heard testimony from survivors

survivors n.
幸存者

legislation n.
立法
American legislation

round-the-clock adj.
全天候

compulsory adj.
强制性的
radio watch compulsory

delegates n.
代表

convention n.
公约
Safety of Life at Sea convention

stipulating v.
规定
stipulating lifeboat seats for all

continuous adj.
不间断

patrol n.
巡逻

stalled v.
暂缓搁置了

treaty n.
条约

regardless adv.
不管怎样

doomed adj.
在劫难逃的

outdated adj.
过时的

stayed afloat adj.
一直没有沉没

discredit v.
信誉扫地
to discredit them

文章简述

- 沉船：1912年，号称几乎不会沉没的泰坦尼克号在首航中撞上冰山，不足三小时沉没，超过1500人遇难。
- 救生：旧法按船体总吨位、也就是容积来定救生艇数，万吨以上只需16艘；泰坦尼克号配了20艘，虽超过法定要求，1178个座位仍只够船上2200人约一半。
- 漏听：附近加州人号的唯一报务员关机睡觉，电台因此没听到求救；他没有违反当时的规定，因为法律未要求全天值守。
- 改革：1912年美国规定无线电必须全天有人值守；1914年13国签约，要求救生艇容纳全员、连续值守并巡查冰情。
- 教训：一战使公约搁置，冰情巡逻仍启动；灾难表明，沿用旧法规，即使配置合法，也未必足以保障安全。

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法老的诅咒

The Curse of the Pharaoh

法老的诅咒，经得起证据检验吗？

英文原文

On 4 November 1922, in the Valley of the Kings, **labourers engaged** by the archaeologist Howard Carter **uncovered** a **staircase carved into** the rock. On 26 November, with his **wealthy sponsor** Lord Carnarvon beside him, Carter made a small hole in the sealed **doorway** and glimpsed the **glittering** gold of the boy king Tutankhamun. The **spectacular** find **riveted** the world.

Carnarvon sold **exclusive** rights to The Times, a **monopoly** that left **competing** reporters **starved of** news. Many **resorted to speculation** or sought outside **commentators**. In March 1923 the **best-selling** novelist Marie Corelli warned publicly that **dire punishment** awaited anyone who entered a sealed **tomb**. Within a **fortnight**, her words would seem **uncanny**.

That month a **mosquito** bit Carnarvon. He **nicked** the bite with a **razor** while **shaving**, it **turned septic**, and blood poisoning led to **pneumonia**. He died in Cairo on 5 April 1923. Newspapers reported that the city's lights failed **at that instant** and that his dog in England **howled** and died. Yet **blackouts** were almost a daily **occurrence** in Cairo.

No curse **inscription** was ever found in the tomb. In December 2002 a British Medical Journal study traced the **fate** of 44 Westerners in Egypt at the time. The 25 **exposed to** the tomb died at an average age of 70, **versus** 75 for the rest, a **margin** not statistically **significant**. The **mortality** figures lent the **ominous rumour** no **credibility**.

Carnarvon's **untimely demise** had **fuelled** the tale, giving it **impetus**. Carter, the **excavator** who had **unsealed** the **burial** place, died in March 1939, aged 64, almost sixteen years after his **patron**. The legend had left him **conspicuously unharmed**.

中文译文

1922年11月4日，在帝王谷，考古学家霍华德·卡特聘用的工人，发掘出了一段雕刻在岩石上面的阶梯。11月26日，在他那富有的赞助人卡纳冯勋爵的陪同下，卡特在封住的门道上开了一个小孔，并瞥见了少年国王图坦卡蒙闪闪发光的黄金。这一惊人的/壮观的发现，牢牢吸引了全世界的注意力。

卡纳冯将独家的新闻报道权卖给了《泰晤士报》，这种垄断让相互竞争的其他记者严重缺少新闻。许多人（只能）诉诸于猜测，或者是找外界的评论员。1923年3月，畅销小说家玛丽·科雷利公开警告，极其严重的惩罚等待着任何进入封闭陵墓的人。两周之内，她的话就会显得诡异而又邪门。

当月，一只蚊子叮了卡纳冯。他刮胡子时用剃刀划破了叮咬处，伤口感染化脓了，继而血液中毒又引发了肺炎。他于1923年4月5日在开罗去世。报纸报道称，就在那一瞬间，全城（开罗）的灯都熄灭了，而他在英国的狗也嚎叫着死去了。然而，停电在当时的开罗几乎是天天都有的事情。

墓中从未发现过任何诅咒铭文。2002年12月，《英国医学杂志》的一项研究追溯了当时身在埃及的44名西方人的命运。其中曾接触这座陵墓的25人，平均死亡年龄为70岁，相比之下其余的人为75岁；这一差距在统计上并不显著。这些死亡数据让这则不祥的传言毫无可信度。

卡纳冯的过早离世助长了这一传言，并为其推波助澜。卡特，这位启封了这处墓葬的发掘者，于1939年3月去世，终年64岁，距他的资助人去世已近十六年。这个传说偏偏让他毫发无损，这一点格外醒目。

词汇与表达

labourers n.
工人

engaged v.
聘用

uncovered v.
发掘出了

staircase n.
阶梯

carved into v.
雕刻在……上面

wealthy adj.
富有的

sponsor n.
赞助人
his wealthy sponsor Lord Carnarvon

doorway n.
门道

glittering adj.
闪闪发光的

spectacular adj.
惊人的/壮观的

riveted v.
牢牢吸引了……注意力
The spectacular find riveted the world

exclusive adj.
独家的
sold exclusive rights

monopoly n.
垄断
a monopoly

competing adj.
相互竞争的

starved of v.
严重缺少
reporters starved of news

resorted to v.
诉诸于
resorted to speculation

speculation n.
猜测
resorted to speculation

commentators n.
评论员

best-selling adj.
畅销

dire adj.
极其严重的
dire punishment

punishment n.
惩罚

tomb n.
陵墓

fortnight n.
两周
Within a fortnight

uncanny adj.
诡异而又邪门
her words would seem uncanny

mosquito n.
蚊子

nicked v.
划破了

razor n.
剃刀

词汇与表达（续）

shaving n.
刮胡子

turned septic adj.
感染化脓了
it turned septic

pneumonia n.
肺炎
blood poisoning led to pneumonia

at that instant n.
在那一瞬间

howled v.
嚎叫

blackouts n.
停电

occurrence n.
事情

inscription n.
铭文
No curse inscription

fate n.
命运

exposed to adj.
接触
The 25 exposed to the tomb

versus prep.
相比之下

margin n.
差距
a margin not statistically significant

significant adj.
显著
not statistically significant

mortality n.
死亡
The mortality figures

ominous adj.
不祥的

rumour n.
传言

credibility n.
可信度
no credibility

untimely adj.
过早

demise n.
离世
Carnarvon's untimely demise

fuelled v.
助长了
had fuelled the tale

impetus n.
推波助澜
giving it impetus

excavator n.
发掘者

unsealed v.
启封了

burial n.
墓葬

patron n.
资助人

conspicuously adv.
格外醒目
conspicuously unharmed

unharmed adj.
毫发无损

文章简述

- 发现：1922年，考古学家卡特与赞助人卡纳冯打开图坦卡蒙墓，墓中珍宝震惊世界。
- 传言：卡纳冯把独家报道权卖给《泰晤士报》，其他记者转向猜测；次年他突然去世，进入墓葬会遭诅咒的故事由此盛行。
- 病因：卡纳冯剃须时划破蚊子叮咬处，伤口感染恶化，并发肺炎；报纸还说他死时开罗停电，但当地本就几乎天天停电。
- 对照：2002年一项研究比较开墓时身在埃及的44人的寿命，接触过墓的25人平均活到70岁，其余未接触者75岁；这点差异不足以排除偶然。
- 结论：墓中从未发现诅咒铭文；卡纳冯去世后，亲自开墓的卡特又活了近16年，现有证据不能证明存在诅咒。

孩子们发明的语言

The Language the Children Invented

没有共同手语的孩子，怎样创造出一门语言？

英文原文

On a **rattling** bus in Managua, deaf pupils kept up a **brisk**, **voiceless hubbub** across the **aisle**, **palms fluttering**, **wrists flicking**. Their teachers could not follow a word.

Before the late 1970s Nicaragua had no deaf **community**. Most deaf people led **secluded** lives, sharing a **homemade repertoire** of gestures with parents and **siblings**. In 1977 a **special-education** centre opened in Managua's San Judas **district** with 50 deaf **pupils**. By 1983 the centre and a **vocational school** for **teenage** pupils held more than 400.

The **curriculum hinged on oral** Spanish and **lipreading**, and teachers signed only the **alphabet**. Pupils studied **lips in vain**, and most failed to **grasp** what a Spanish word was. **At recess**, in the street, and on the buses, they **mingled** gestures **with** home signs. A **rough jumble ripened rapidly into** a **coherent** language **entirely** their own.

The teachers found it **unintelligible**, and in June 1986 **the Ministry of Education brought in** Judy Kegl, an **American Sign Language linguist** from **MIT**. A 2001 study found that pupils who arrived younger, and later **intakes**, used **grammatical** devices more often and more **uniformly** than their **forerunners**. Each fresh wave of young **signers** added **complexity**.

In a 2004 study, gesturing Spanish speakers and the earliest signers described something **rolling downhill** by **welding** how it moved and where it went **into** one **undivided** movement. Later signers **dismantled** it **into discrete segments** in **linear order**, a **framework** found **nowhere** in the gestures around them.

英文原文（续）

Linguists had a rare chance to watch a language grow from **infancy** to **maturity**. Its **builders** were not teachers or experts but deaf **playmates** in playgrounds, streets, and buses.

中文译文

在一辆哐当作响的马那瓜市的公车上，聋人学生们一直进行着一场轻快的、无声的喧闹，隔着过道，手掌拍动着，手腕轻弹。他们的老师一个字也看不懂。

1970年代末以前，尼加拉瓜没有聋人群体。大多数聋人过着与世隔绝的生活，共用自制的一整套/全部手势，在父母和兄弟姐妹之间。1977年，一所特殊教育中心开办于马那瓜的圣胡达斯区，共有50个聋哑学生。到1983年，这所中心和一所面向青少年学生的职业学校，共有超过400多个聋哑学生。

课程以西班牙语口语和唇读为中心，并且老师们只用手语比划字母（表）。学生们仔细观察着唇语，却是徒劳无功，多数人没能弄明白/理解什么是一个西班牙语单词。在课间休息时，在街上，以及在公共汽车上，他们混合了手势与自家手势。一堆粗糙的杂乱的东西迅速地成熟为一门连贯的语言，完全属于他们自己。

老师们发现这种手语很难理解，并且在1986年6月，教育部请来了朱迪·凯格尔，一位美国手语语言学家，来自麻省理工学院。2001年的一项研究发现，入学年龄更小的学生，以及后来招收的几批（学生），使用语法手段更频繁，也更统一，比起他们的前辈。每一批新的年幼手语者都为这门语言增添了复杂性。

在2004年的一项研究中，打手势的西班牙语使用者和最早一批手语者描述某个东西滚下坡是通过把它怎样运动和往哪里去熔接成一个完整的/统一的动作。后来的手语者把它拆解成离散的片段，按线性顺序排列，这种框架/结构在他们周围的手势中无处可寻。

语言学家难得有机会看着一门语言从婴儿期长到成熟期。它的建造者不是老师或专家，而是聋哑人的玩伴们，他们在游乐场，在街头，以及在巴士上。

词汇与表达

rattling adj.
哐当作响的

brisk adj.
轻快的

voiceless adj.
无声的

hubbub n.
喧闹

aisle n.
过道

palms n.
手掌

fluttering v.
拍动着

wrists n.
手腕

flicking v.
轻弹

community n.
群体
no deaf community

secluded adj.
与世隔绝的
led secluded lives

homemade adj.
自制的

repertoire n.
一整套/全部
a homemade repertoire of gestures

siblings n.
兄弟姐妹

special-education adj.
特殊教育

district n.
区

pupils n.
学生

vocational school adj.
职业学校

teenage adj.
青少年

curriculum n.
课程
The curriculum hinged on oral Spanish

hinged on v.
以……为中心
hinged on oral Spanish and lipreading

词汇与表达（续）

oral adj.
口语

lipreading n.
唇读

alphabet n.
字母（表）

lips n.
唇语

in vain n.
徒劳无功
Pupils studied lips in vain

grasp v.
弄明白/理解
failed to grasp

At recess n.
在课间休息时

mingled with v.
混合了……与

rough adj.
粗糙的

jumble n.
杂乱的东西

ripened into v.
成熟为

rapidly adv.
迅速地

coherent adj.
连贯的
a coherent language

entirely adv.
完全

the Ministry of Education n.
教育部

unintelligible adj.
很难理解
The teachers found it unintelligible

brought in v.
请来了

American Sign Language n.
美国手语

linguist n.
语言学家
an American Sign Language linguist

MIT n.
麻省理工学院

intakes n.
招收的几批
later intakes

词汇与表达（续）

grammatical adj.
语法
used grammatical devices

uniformly adv.
统一
more uniformly than their forerunners

forerunners n.
前辈

signers n.
手语者

complexity n.
复杂性
added complexity

rolling downhill adv.
滚……下坡
described something rolling downhill by welding how it moved

welding into v.
把……熔接成

undivided adj.
完整的/统一的

dismantled into v.
拆解……成

discrete adj.
离散的
discrete segments in linear order

segments n.
片段
discrete segments in linear order

linear order adj.
线性顺序
segments in linear order

framework n.
框架/结构
a framework found nowhere

nowhere adv.
无处

infancy n.
婴儿期
from infancy to maturity

maturity n.
成熟期
from infancy to maturity

builders n.
建造者

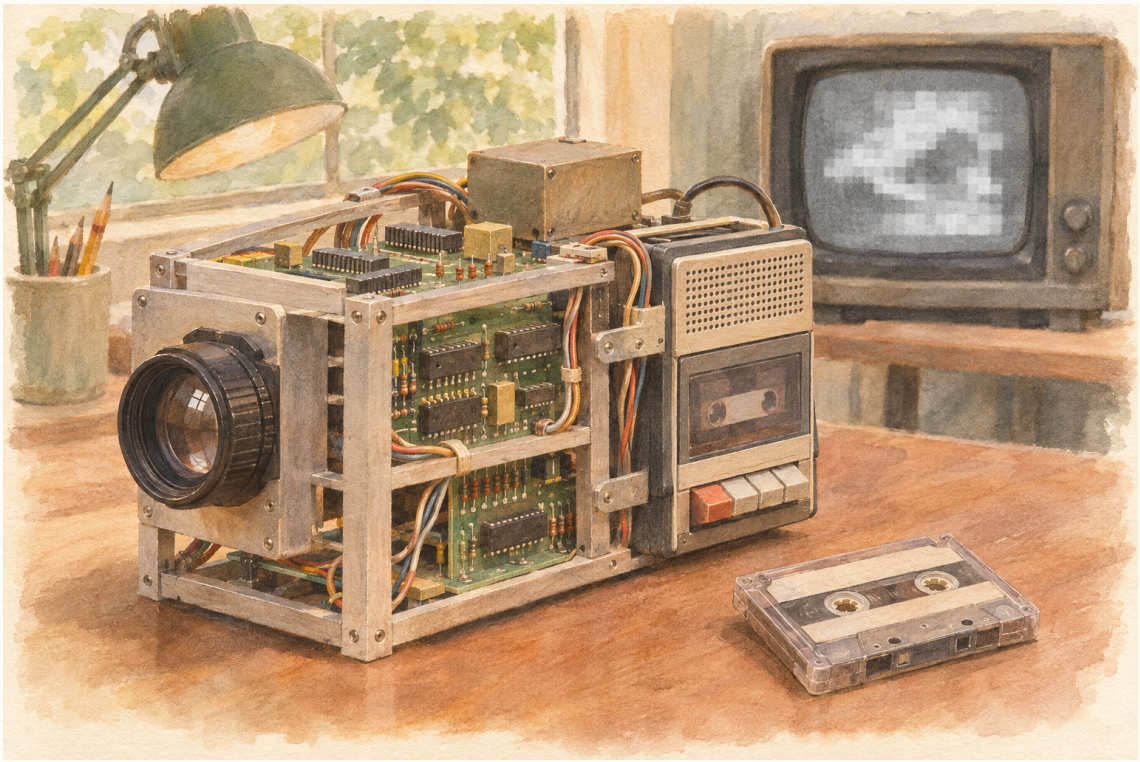
playmates n.
玩伴们

文章简述

- 相聚：1977年，尼加拉瓜的学校让原本彼此隔绝的聋童聚到一起。此前，他们大多只与家人用自创手势交流。
- 受阻：学校教西班牙语口语和看口形，多数孩子学不会；课外，他们却在操场、街头和公车上用手势交流。
- 诞生：孩子们把各家的手势混合起来，逐渐形成一门大家都能使用、老师却看不懂的新语言。
- 成长：年龄更小和后来入学的孩子，让表达规则更统一、更复杂。例如说“滚下坡”，早期用一个动作一起比划，后来会把“怎么动”和“往哪去”分开，按顺序表达。
- 意义：这门语言是在孩子们的同伴交流中发展出来的，语言学家因而有机会亲眼观察一门语言逐步成熟。

发明了数码相机，却被它打败

The Company That Invented Its Own Killer



造出数码相机的公司，为什么仍被数码时代冲击？

英文原文

In 1975 Kodak engineer Steven Sasson built the first **digital camera**. The **toaster**-sized **gadget**, a **cumbersome heavyweight** of about eight pounds, was hardly **portable**, **let alone handheld**. Its first **rudimentary** picture held 10,000 black-and-white **pixels**. Recording it on a **cassette** took 23 seconds, and a **playback** unit showed it on a television.

Kodak's **revenues** showed its **dependence on** film, the **profitable pillar** of its business. Technical **staffers** were **appreciative of** the camera, but it needed no film. By Sasson's **recollection**, his **superiors** at **headquarters** called it cute and urged him to **keep it hushed**. A **patent** followed in 1978. Production never did.

The **advent** of digital photography proved **disruptive**. In June 2009 Kodak said it would stop making Kodachrome, its **analogue** film, citing **lessened** demand in the digital **era**. Kodak's **manpower** had reached an **apex** of about 145,000 in the 1980s. By 2012, **layoffs** since 2003 had **slashed** 47,000 jobs, alongside the **shutdown** of 13 factories and some 130 labs.

On 19 January 2012 the 131-year-old firm sought Chapter 11 protection from its **creditors**, **sapped** by **pension** costs and years of **slowness** in going digital. Citigroup, a **lender**, provided $950 million for its **restructuring**. A **portfolio** of 1,100 digital imaging patents fetched about $525 million from a **consortium** of twelve **buyers**, including the **giants** Apple and Google.

After 19 months of **reorganisation**, the **once-mighty** Kodak left bankruptcy in September 2013, **downsized**, **streamlined**, and **refocused as** a commercial **printer** for business **clients**. In November 2009 President Obama had **decorated** Sasson with national **laurels**, **commending** his digital camera. That **medal**'s year also saw Kodak announce **the end of** Kodachrome.

中文译文

1975年，柯达工程师史蒂文·萨松造出了第一台数码相机。这台烤面包机大小的小器械/小装置，是个笨重的大块头，重约八磅，很难说是便携的，更谈不上手持。它拍下的第一张粗糙的照片，只有1万个黑白像素。把照片录到盒式磁带上要花23秒，再由一台回放装置在电视上显示出来。

柯达的收入显示出公司对胶卷的依赖，胶卷是其业务中赚钱的支柱。技术员工对这台相机颇为赞赏，但它不需要胶卷。据萨松回忆，他在总部的上司说它挺可爱，并敦促他对此秘而不宣。1978年，这台相机获得专利，生产部门却从未这样做过（跟进）。

数码摄影的问世，结果证明是颠覆性的。2009年6月，柯达宣布将停产Kodachrome这种模拟胶卷，理由是数码时代里需求减少。柯达的员工规模曾达到约14.5万人的顶峰，那是在20世纪80年代。到2012年，自2003年以来的裁员已砍掉4.7万个岗位，同时13家工厂和约130个冲印室相继关闭。

2012年1月19日，这家已有131年历史的公司申请《破产法》第11章保护，以免受债权人追讨，它受养老金负担和多年来转向数码的迟缓所慢慢抽空/消耗。贷款方（借出钱的人）花旗集团提供了9.5亿美元，用于其重组。由1,100项数字成像专利构成的组合以约5.25亿美元卖给一个由十二家买家组成的财团，其中包括巨头苹果和谷歌。

经过19个月的改组，昔日强大的柯达于2013年9月走出破产，规模已缩小，业务经过精简，重新定位为商业印刷商，面向企业客户。2009年11月，奥巴马总统授予萨松国家级荣誉，来表彰他的数码相机。那枚奖章颁发的同年，柯达也宣布了Kodachrome 的终结。

词汇与表达

digital camera n.
数码相机

toaster n.
烤面包机

gadget n.
小器械/小装置

cumbersome adj.
笨重的
a cumbersome heavyweight

heavyweight n.
大块头

portable adj.
便携的

let alone conj.
更谈不上

handheld adj.
手持

rudimentary adj.
粗糙的
Its first rudimentary picture

pixels n.
像素

cassette n.
盒式磁带

playback n.
回放

revenues n.
收入
Kodak's revenues

dependence on n.
对……依赖
its dependence on film

profitable adj.
赚钱的
the profitable pillar of its business

pillar n.
支柱

staffers n.
员工

appreciative of adj.
赞赏
appreciative of the camera

recollection n.
回忆
By Sasson's recollection

superiors n.
上司

headquarters n.
总部

keep it hushed adj.
秘而不宣
urged him to keep it hushed

patent n.
专利
A patent followed in 1978

advent n.
问世
The advent of digital photography

disruptive adj.
颠覆性的
proved disruptive

analogue adj.
模拟

lessened adj.
减少

词汇与表达（续）

era n.
时代

manpower n.
员工规模

apex n.
顶峰

layoffs n.
裁员
layoffs since 2003

slashed v.
砍掉

shutdown n.
关闭

creditors n.
债权人
protection from its creditors

sapped v.
慢慢抽空/消耗
sapped by pension costs

pension n.
养老金

slowness n.
迟缓

lender n.
贷款方（借出钱的人）

restructuring n.
重组
for its restructuring

portfolio n.
组合
A portfolio of 1,100

consortium n.
财团
a consortium of twelve buyers

buyers n.
买家

giants n.
巨头

reorganisation n.
改组
19 months of reorganisation

once-mighty adj.
昔日强大的

downsized adj.
缩小
downsized, streamlined

streamlined adj.
精简
downsized, streamlined

refocused as v.
重新定位为
refocused as a commercial printer

printer n.
印刷商

clients n.
客户

decorated v.
授予

laurels n.
荣誉

commending v.
表彰
commending his digital camera

medal n.
奖章

the end of n.
终结

文章简述

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- 牵制：胶卷是利润支柱；据萨松回忆，上司让他别声张。原型机1978年获专利，却未投产。
- 衰退：数码摄影冲击胶卷需求，转型迟缓与养老金负担削弱公司，柯达于2012年申请破产保护。
- 重组：柯达获9.5亿美元贷款，并以约5.25亿美元出售1100项专利；2013年完成重组，转做企业印刷。
- 反差：2009年，萨松因数码相机获美国国家荣誉；同年，柯达宣布停产Kodachrome这一胶卷产品。

青蒿一握，济世千万

A Handful of Wormwood That Saved Millions

古方里没有煮沸这一步，怎样启发了青蒿素研究？

英文原文

Project 523, a **covert military mission** involving many institutes, was named after its launch date, 23 May 1967. Its **urgent** target was the chloroquine-**resistant malaria weakening troops** in the **Vietnam War**. In January 1969 Tu Youyou was **entrusted with** leading her institute's research group, **despite** having no **doctorate** or foreign training.

In an **exhaustive** search, her team **combed** over 2,000 **folk** recipes, **compiled** 640 **prescriptions**, and **evaluated** about 380 **extracts** from some 200 **medicinal herbs**. Qinghao, or **sweet wormwood**, showed **flashes** of **promise**, yet its results were **erratic**. Seeking a **clue**, Tu **revisited** the **classical** texts.

Around AD 340, the physician Ge Hong's **handbook** advised taking "a handful of qinghao **immersed in** two **litres** of water". This **bypassed** boiling. Tu suspected heat during **extraction** was **degrading** a **delicate component**, so she **switched to ether**, a **volatile solvent** boiling at about 35 **degrees Celsius**. On 4 October 1971 sample 191 completely **inhibited** the **parasite** in **infected** mice.

Efficacy in mice **indicated** nothing about human safety. In July 1972, to **assess** its **toxicity**, Tu and two colleagues **voluntarily** took the extract. **Thereafter**, from August to October, it was **administered to** 21 malaria **patients** in Hainan, and on 8 November her team **obtained** pure **crystals** of the **substance** later named artemisinin.

In September 2011 Tu was named a Lasker Award **recipient**; its **jury** credited the drug with saving millions. On 5 October 2015 she won half the Nobel Prize in **Physiology** or Medicine, the first Chinese woman Nobel **laureate**. Since 2006 WHO **guidelines** have **endorsed** artemisinin-based **combination therapies** against malaria's **deadliest** form. The **essence** of Ge Hong's line lay in the boiling it **skipped**.

中文译文

523项目是一项秘密的/隐秘的军事任务，涉及多家机构，得名于它的启动日期，1967年5月23日。它的紧迫目标是对氯喹产生耐药性的疟疾，它削弱了越南战争中的部队。1969年1月，屠呦呦受命领导她所在研究所的研究组，尽管她既没有博士学位，也没有在国外受过训练。

在一次彻底的搜寻中，她的团队仔细翻检了2,000多个民间药方，编纂/汇编了640个处方/药方，并评估了约380种提取物，它们来自200种左右的药用的草药。青蒿，也就是黄花蒿，展现出几丝成功迹象的闪光/闪现，然而它的效果却并不稳定/时好时坏的。为了寻找线索，屠呦呦重新审阅了古典文献。

约公元340年，医师葛洪的手册建议服用：“一把浸泡在两升水中的青蒿”。这个方法绕开了煮沸。屠呦呦怀疑，提取过程中的高温正在降解/分解某种脆弱的成分，于是她改用/换成了乙醚，一种易挥发的溶剂，沸点为约35摄氏度。1971年10月4日，第191号样品完全抑制了受感染小鼠体内的寄生虫。

在小鼠身上有药效，并不能说明人是否安全。1972年7月，为了评估它的毒性，屠呦呦和两名同事自愿地服下了这种提取物。此后，从8月到10月，这种提取物用于治疗21名疟疾病人，在海南，而在11月8日，她的团队得到了这种物质的纯净晶体，它后来被命名为青蒿素。

2011年9月，屠呦呦被评为拉斯克奖得主；该奖的评审团将挽救了数百万人的功劳归功于这种药物。2015年10月5日，她获得了一半的诺贝尔生理学或医学奖，成为第一位中国女性诺贝尔奖获得者。自2006年起，世界卫生组织的指南一直公开支持/赞同以青蒿素为基础的联合疗法，用来对付疟疾中最致命的一种。葛洪那句话的精髓，正在于它略去的煮沸。

词汇与表达

covert adj.
秘密的/隐秘的

military adj.
军事

mission n.
任务

urgent adj.
紧迫

resistant adj.
产生耐药性的
chloroquine-resistant malaria

malaria n.
疟疾

weakening v.
削弱了

troops n.
部队

Vietnam War n.
越南战争

entrusted with v.
受命
entrusted with leading

despite prep.
尽管

doctorate n.
博士学位

exhaustive adj.
彻底的
In an exhaustive search

combed v.
仔细翻检了
combed over 2,000 folk recipes

folk adj.
民间

compiled v.
编纂/汇编了
compiled 640 prescriptions

prescriptions n.
处方/药方

evaluated v.
评估了
evaluated about 380 extracts

extracts n.
提取物
evaluated about 380 extracts

medicinal adj.
药用的

herbs n.
草药

词汇与表达（续）

sweet wormwood n.
黄花蒿

flashes n.
闪光/闪现

promise n.
成功迹象

erratic adj.
并不稳定/时好时坏的
its results were erratic

clue n.
线索

revisited v.
重新审阅了

classical adj.
古典

handbook n.
手册

immersed in v.
浸泡在

litres n.
升

bypassed v.
绕开了

extraction n.
提取
heat during extraction

degrading v.
降解/分解
was degrading a delicate component

delicate adj.
脆弱的

component n.
成分
a delicate component

switched to v.
改用/换成了

ether n.
乙醚

volatile adj.
易挥发的
a volatile solvent

solvent n.
溶剂
a volatile solvent

degrees Celsius n.
摄氏度

inhibited v.
抑制了
completely inhibited the parasite

parasite n.
寄生虫

infected adj.
受感染

词汇与表达（续）

Efficacy n.
药效
Efficacy in mice

indicated v.
说明

assess v.
评估
to assess its toxicity

toxicity n.
毒性
to assess its toxicity

voluntarily adv.
自愿地

Thereafter adv.
此后

administered to v.
用于治疗
administered to 21 malaria patients

patients n.
病人

obtained v.
得到了

crystals n.
晶体

substance n.
物质
pure crystals of the substance

recipient n.
得主

jury n.
评审团

Physiology n.
生理学

laureate n.
获得者

guidelines n.
指南

endorsed v.
公开支持/赞同
have endorsed artemisinin-based

combination n.
联合
combination therapies

therapies n.
疗法

deadliest adj.
最致命的

essence n.
精髓
The essence of Ge Hong's line

skipped v.
略去

文章简述

- 寻药：部分疟疾已对旧药氯喹产生抗药性，中国启动523联合研究项目寻找新药。屠呦呦团队查古方、测试多种草药提取物。
- 线索：青蒿有时有效、有时无效。葛洪古方只说浸泡、不说煮沸，屠呦呦怀疑提取时的高温破坏了有效成分。
- 突破：团队改用低温乙醚提取，1971年的样品完全抑制了感染小鼠体内的疟原虫，但小鼠有效仍不能证明人服用安全。
- 验证：1972年，她与两名同事先试服，随后给21名疟疾患者使用；团队又分离出纯晶体，后来命名为青蒿素。
- 影响：以青蒿素为基础、与其他药物合用的疗法获世卫推荐，挽救众多生命；屠呦呦于2015年获诺贝尔奖。

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洞穴里的十三个人

Thirteen in the Cave

找到十三人之后，为什么救他们出洞仍如此困难？

英文原文

On 23 June 2018, twelve boys aged 11 to 16 from a **junior** football **squad went on an outing** with their 25-year-old **assistant coach**. They **entered** Tham Luang, a **sprawling maze** of **tunnels** in northern Thailand, to **penetrate** its **gloom**. **Monsoon** rain then **flooded** the **passages**, and the rising water trapped all of them.

For nine days their **whereabouts** were unknown. On 2 July two **elite** British **divers**, **threading through submerged passages**, reached an **inundated** chamber about 4 km from the **entrance**. All 13 were there, **huddled** on a **muddy mound** and **perched** just above the water, every one of them **alive**.

The only **exit ran through kilometres** of **constricted**, **treacherous passages**, many of them **navigable** only with **scuba** gear. **Rescuers** began **stockpiling** oxygen **cylinders along the route**. On 6 July Saman Kunan, a **former** Thai Navy SEAL, fell **unconscious** while laying them and died.

The **evacuation** began on 8 July, and the boys came out **in batches**. Each was given **doses** of **sedative**, **wrapped motionless** in a **stretcher**, and passed from diver to diver, with **medics stationed along the route**. On 10 July the last four **footballers** and their coach were **escorted** out after their **teammates**.

The **multinational mobilisation** was a feat of **logistics**. Some 10,000 people **converged on** the site, including a **contingent** of more than 100 divers, plus 2,000 soldiers and 900 **policemen**, with teams **deployed** from abroad. The rescue had a **sobering postscript**: in December 2019 another Navy SEAL died of a **blood infection** caught during the operation. Every boy came home alive, and two rescuers were **mourned**.

中文译文

2018年6月23日，12名男孩，年龄在11至16岁，来自一支青少年足球小队，外出郊游，同行的是他们25岁的助理教练。他们进入了睡美人洞，那是一座由隧道构成、向四周分散延伸的迷宫，位于泰国北部；他们想要穿透/深入洞中的幽暗。季风带来的雨水随后淹没了通道，而上涨的水把他们全部困住了。

九天里，他们的下落不明。7月2日，两名英国精英潜水员穿过水下通道，到达一个被淹的洞室，距离洞穴的入口大约4公里。13人都在那里，挤在一起在一处泥泞的土堆上，刚好栖身于水面之上，他们每个人都还活着。

唯一的出口要穿过数公里狭窄且暗藏危险的通道，其中许多段只有借助水肺装备才能通行。救援人员开始储备氧气钢瓶，放在沿途。7月6日，萨曼·古南，一名前泰国海军海豹突击队员，在布放钢瓶时失去知觉，随后死亡。

撤离开始于7月8日，男孩们分批出洞。每人都用了数剂镇静药，一动不动地包裹在担架里，由潜水员一个接一个传递出去，伴随着医护人员驻守在沿途。7月10日，最后四名小球员和教练继队友之后被护送出洞。

这场多国的动员被堪称为一场后勤壮举。约1万人汇集到现场，其中包括一支100多名潜水员组成的分队，另有2,000名士兵和900名警察，还有一些救援小组，从国外调派而来。这次救援留下了一个令人警醒的后续：2019年12月，又一名海军海豹突击队员死于在救援中染上的血液感染。每个男孩都活着回到了家，而两名救援者却只能永远被悼念。

词汇与表达

junior adj.
青少年

squad n.
小队

went on an outing v.
外出郊游

assistant coach n.
助理教练

entered v.
进入了

sprawling adj.
向四周分散延伸的
a sprawling maze of tunnels

maze n.
迷宫
a sprawling maze of tunnels

tunnels n.
隧道

penetrate v.
穿透/深入

gloom n.
幽暗

Monsoon n.
季风

flooded v.
淹没了

passages n.
通道

whereabouts n.
下落
their whereabouts were unknown

elite adj.
精英

divers n.
潜水员

threading through v.
穿过
threading through submerged passages

submerged adj.
水下
submerged passages

inundated adj.
被淹的
an inundated chamber

entrance n.
入口

huddled v.
挤在一起

词汇与表达（续）

muddy adj.
泥泞的

mound n.
土堆

perched v.
栖身

alive adj.
活着

exit n.
出口

ran through v.
穿过

kilometres n.
公里

constricted adj.
狭窄
constricted, treacherous passages

treacherous adj.
暗藏危险的
constricted, treacherous passages

navigable adj.
通行
navigable only with scuba gear

scuba n.
水肺

Rescuers n.
救援人员

stockpiling v.
储备
stockpiling oxygen cylinders

cylinders n.
钢瓶

along the route prep.
沿途

former adj.
前

unconscious adj.
失去知觉
fell unconscious

evacuation n.
撤离
The evacuation began on 8 July

in batches n.
分批
the boys came out in batches

doses n.
数剂

词汇与表达（续）

sedative n.
镇静药
doses of sedative

wrapped v.
包裹

motionless adj.
一动不动地
wrapped motionless in a stretcher

stretcher n.
担架

medics n.
医护人员

stationed v.
驻守

footballers n.
小球员

escorted v.
护送
were escorted out

teammates n.
队友

multinational adj.
多国的
The multinational mobilisation

mobilisation n.
动员
The multinational mobilisation

logistics n.
后勤
a feat of logistics

converged on v.
汇集到

contingent n.
分队

policemen n.
警察

deployed v.
调派
teams deployed from abroad

sobering adj.
令人警醒的
a sobering postscript

postscript n.
后续

blood infection n.
血液感染

mourned v.
悼念
two rescuers were mourned

文章简述

- 被困：2018年，泰国12名少年球员和助理教练进入睡美人洞，因季风雨淹没通道而受困。
- 发现：失联九天后，两名英国潜水员在距洞口约4公里处找到他们，十三人全部活着。
- 难题：救援需穿过数公里狭窄水道；约万人的多国队伍沿途布放气瓶，协调百余名潜水员。
- 救出：7月8日至10日，男孩用镇静药后固定在担架中，由潜水员接力带出，沿途有医护协助。
- 代价：十三人全获救；救援者萨曼·古南在前期布气瓶时遇难，另一名救援队员2019年死于救援时染上的血液感染。